Figures that preserve evidence

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

One complete research-case evidence record.

Preserve units, case counts and uncertainty limits without making a figure stand in for missing evidence.

Keep one habit: Recall → W1–W3/W5 → independent case → project evidence → feedback and delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Negative results and claim revision. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Power and resource tradeoffs. Give an example and a mistake that the idea helps you avoid.

Negative results and claim revision

My remembered idea, example and correction

Inspecting failures and anomalies

My remembered idea, example and correction

Power and resource tradeoffs

My remembered idea, example and correction

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Figures represent evidence and must preserve units, denominators, scales and uncertainty limits. Main supplied rates A 80%, B 82% differ by two percentage points, or 2.5% relative to A. Raw case counts are unknown; do not invent eighty or eighty-two cases out of one hundred. Bars starting secretly at 79% have visible heights one and three, exaggerating the actual ratio.

  Use zero-baseline magnitude bars or a clearly explained difference plot. Separate picture has A 6/10 = 60%, B 8/10 = 80%, difference two correct or twenty percentage points. A hidden start at five gives drawn heights one/three and misleads. Keep ten-case scope visible. No error bars or significance information are supplied.

  Unequal-denominator counts require rates and both totals, plus case/eligibility context; rate ordering alone does not establish a controlled or causal advantage. A polished chart does not improve sampling or add observations.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the worked case

> Keep the supplied inputs, comparison rule and evidence together.

Complete task and evidence

- Figures represent evidence and must preserve units, denominators, scales and uncertainty limits. Main supplied rates A 80%, B 82% differ by two percentage points, or 2.5% relative to A. Raw case counts are unknown; do not invent eighty or eighty-two cases out of one hundred. Bars starting secretly at 79% have visible heights one and three, exaggerating the actual ratio.

  Use zero-baseline magnitude bars or a clearly explained difference plot. Separate picture has A 6/10 = 60%, B 8/10 = 80%, difference two correct or twenty percentage points. A hidden start at five gives drawn heights one/three and misleads. Keep ten-case scope visible. No error bars or significance information are supplied.

  Unequal-denominator counts require rates and both totals, plus case/eligibility context; rate ordering alone does not establish a controlled or causal advantage. A polished chart does not improve sampling or add observations.

- Rates A **80%**, B 82% are plotted on bars starting 79%, making B look several times larger. Show the intermediate steps and give the conclusion with its scope.

Main figure calculation	Complete value	Meaning
A rate	80%	Supplied rate; underlying denominator unknown
B rate	82%	Supplied rate; underlying denominator unknown
Absolute difference	82% – 80% = 2 percentage points	Correct rate-scale difference
Relative increase	2/80 = 2.5%	Different from two percentage points
Heights above hidden 79% baseline	A: 1 point; B: 3 points	Threefold drawn heights distort the actual rate ratio

Main figure calculation	Complete value	Meaning
Repair	Zero-baseline percentage bars or clearly labelled difference plot	Retain missing case-count and uncertainty information

My reasoning and evidence

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Figures represent evidence and must preserve units, denominators, scales and uncertainty limits. Main supplied rates A 80%, B 82% differ by two percentage points, or 2.5% relative to A. Raw case counts are unknown; do not invent eighty or eighty-two cases out of one hundred. Bars starting secretly at 79% have visible heights one and three, exaggerating the actual ratio.

  Use zero-baseline magnitude bars or a clearly explained difference plot. Separate picture has A 6/10 = 60%, B 8/10 = 80%, difference two correct or twenty percentage points. A hidden start at five gives drawn heights one/three and misleads. Keep ten-case scope visible. No error bars or significance information are supplied.

  Unequal-denominator counts require rates and both totals, plus case/eligibility context; rate ordering alone does not establish a controlled or causal advantage. A polished chart does not improve sampling or add observations.

- A report omits this qualification: Visual choices must reflect the intended metric and cannot imply causal certainty from association.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Figures represent evidence and must preserve units, denominators, scales and uncertainty limits. Main supplied rates A 80%, B 82% differ by two percentage points, or 2.5% relative to A. Raw case counts are unknown; do not invent eighty or eighty-two cases out of one hundred. Bars starting secretly at 79% have visible heights one and three, exaggerating the actual ratio.

  Use zero-baseline magnitude bars or a clearly explained difference plot. Separate picture has A 6/10 = 60%, B 8/10 = 80%, difference two correct or twenty percentage points. A hidden start at five gives drawn heights one/three and misleads. Keep ten-case scope visible. No error bars or significance information are supplied.

  Unequal-denominator counts require rates and both totals, plus case/eligibility context; rate ordering alone does not establish a controlled or causal advantage. A polished chart does not improve sampling or add observations.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Counts A 8/10, B 9/15 are plotted as 8 versus 9 without denominators. Correct.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the research evidence

> Preserve the question, protocol, observations, analysis and limits.

Complete task and evidence

- Add a labelled figures that preserve **evidence evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of figures that preserve **evidence**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Figures that preserve **evidence**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Figures that preserve **evidence**. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Counts A 8/10, B 9/15 are plotted as 8 versus 9 without denominators. Correct.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases.

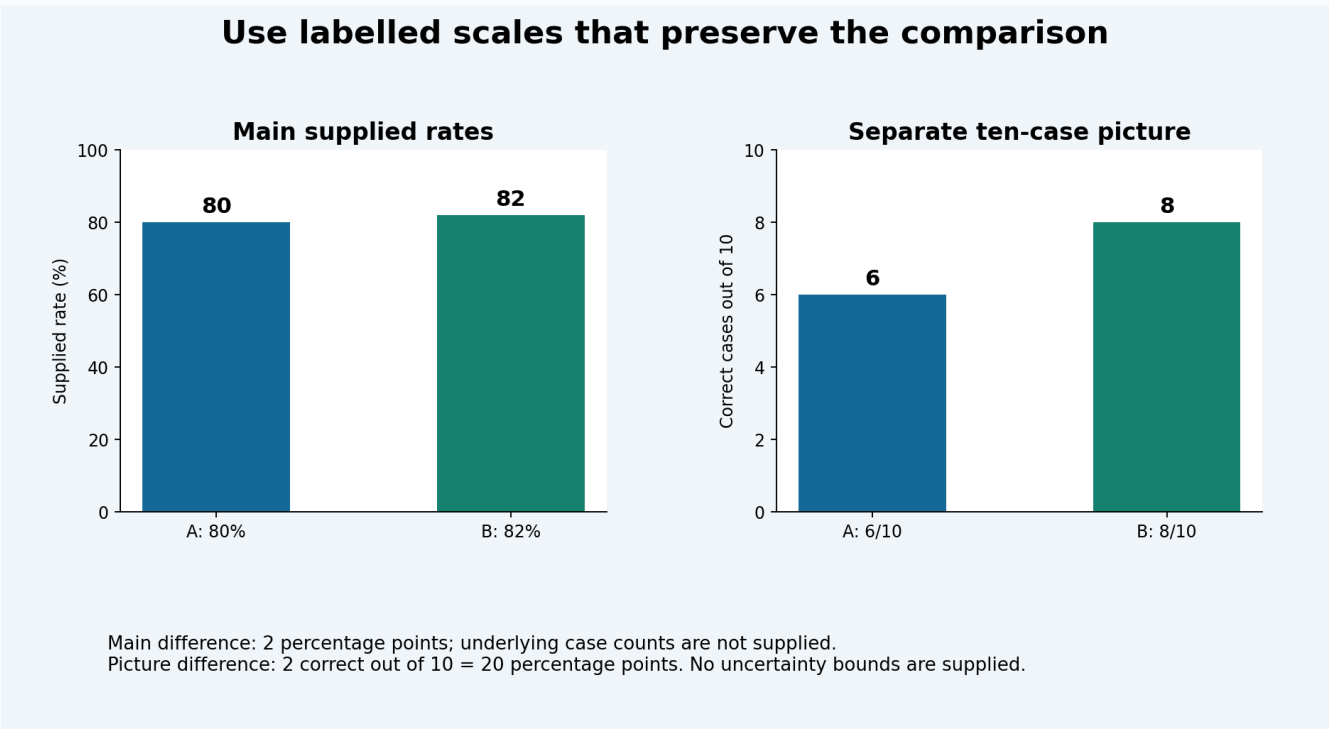
- A has 6/10 correct; B has 8/10. The count axis should start at zero and identify the ten-case denominator.

- Draw count bars for 6/10 and 8/10 using the stated axis. Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?

- Support used: alone / hint / worked example. This record is practice **evidence**.

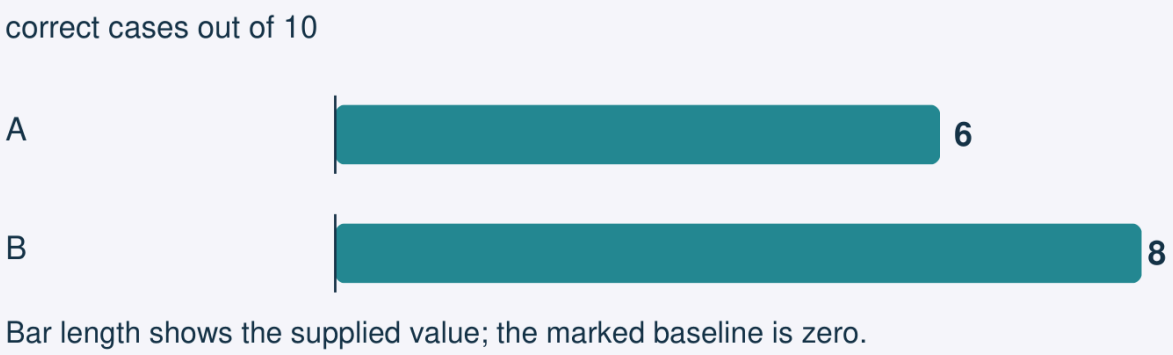
- Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

  Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.



AI ACADEMY / AI-7/29

Zero-based correct counts



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/29

Figures that preserve evidence

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Draw honest bars

Use heights 6 and 8 on a zero-based correct-count axis.



2. Label the scale

State counts out of ten, not an unspecified quality score.



3. Keep context nearby

Note the fixed tasks and absence of a significance test.

CHANGE ONE THING

Start the axis at 5 without making the truncation obvious.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Reproducible AI Research Study
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases.
- Which labelled step matches this detail? Note the fixed tasks and absence of a significance test.
- 1. Draw honest bars
- 2. Label the scale
- 3. Keep context nearby
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Keep context nearby Note the fixed tasks and absence of a significance test. Figures make **evidence** easier to inspect when labels and scales preserve meaning.

A dramatic-looking visual cannot supply missing uncertainty or representativeness.

Changed-case question and optional hint

- Changed-case practice: Start the axis at 5 without making the truncation obvious. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The bars exaggerate the visual contrast and require correction or clear disclosure.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The bars exaggerate the visual contrast and require correction or clear disclosure. Visible heights no longer correspond to the full count scale.

Repair and retain the record

Review Negative results and claim revision (ai-7/28). Goal: Explain the **mechanism** and terminology of negative results and claim revision. Short practice: Does a 6/8 tie support the **prediction** that B scores higher?

Explain why one newly selected favourable demonstration does not replace the original result or prove equivalence.. Return to this week and explain what you changed.

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the **mechanism** and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10.

What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

Review Power and resource tradeoffs (ai-7/23). Goal: Explain the **mechanism** and terminology of power and resource tradeoffs. Short practice: Calculate both simulated detection fractions and resource costs.

What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Figures that preserve evidence: the case

AI ACADEMY / AI-7/29 · 01 · LOOK AT THE FACTS

Figures that preserve evidence: the case

The situation

Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases.

Supplied evidence

A has 6/10 correct; B has 8/10. The count axis should start at zero and identify the ten-case denominator.

Explain figures that preserve evidence using the records and boundaries of this fictional investigation.

Follow the labelled picture with the matching explanation. Open the vector diagram

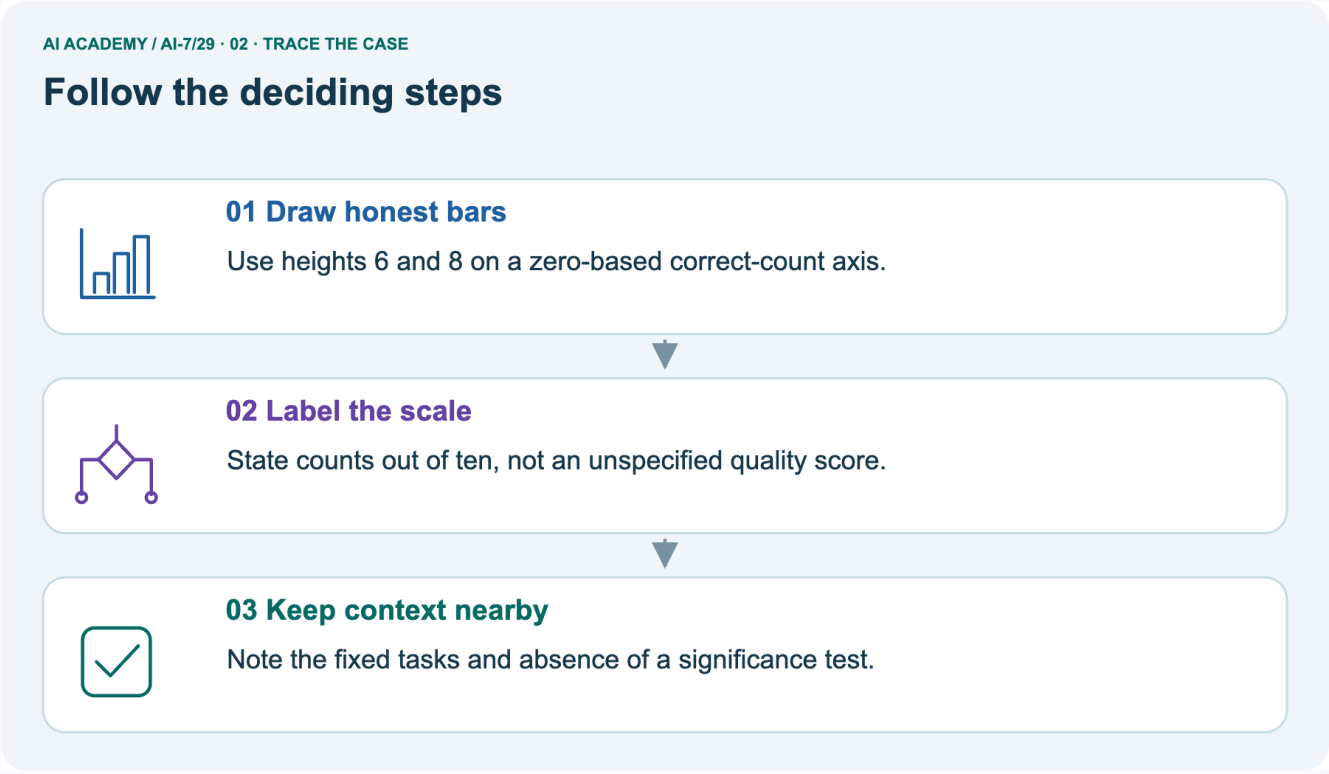
Draw bars that preserve the comparison. A report plots correct counts for two versions on the same ten cases. Supplied facts: A has 6/10 correct; B has 8/10.

The count axis should start at zero and identify the ten-case denominator.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



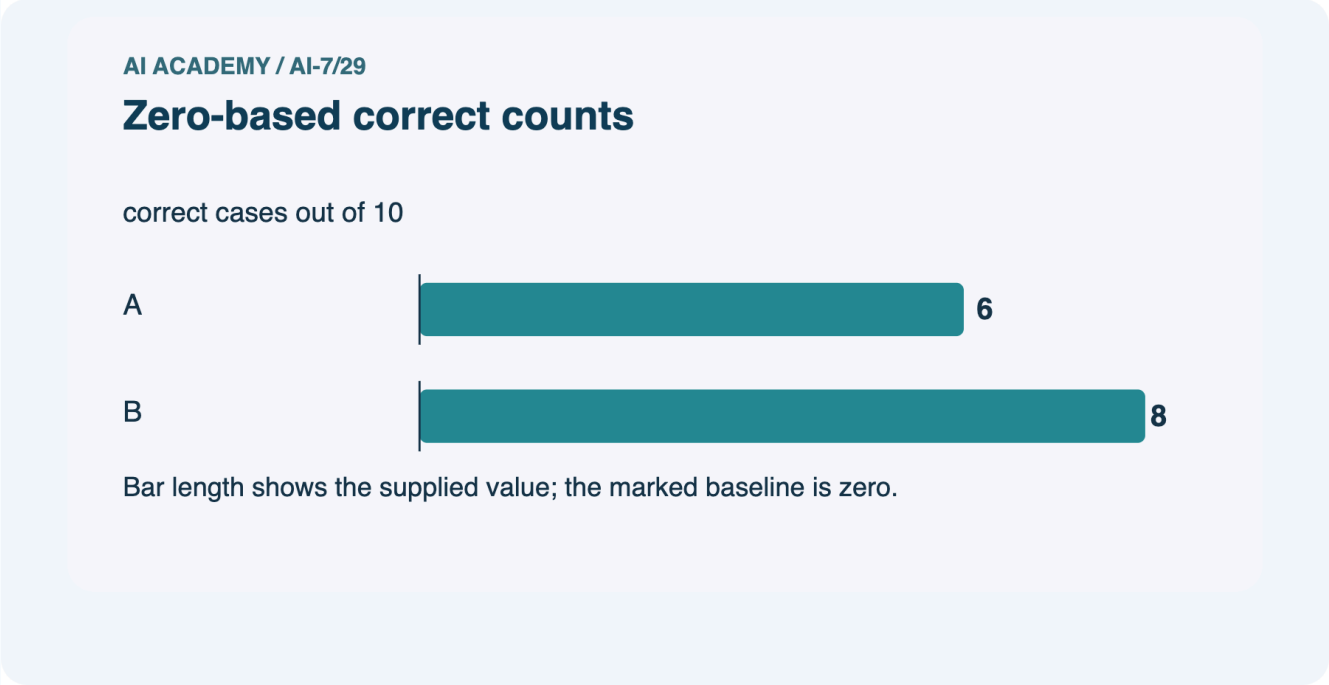
Follow the labelled picture with the matching explanation. Open the vector diagram

Draw honest bars: Use heights 6 and 8 on a zero-based correct-count axis. → Label the scale: State counts out of ten, not an unspecified quality score. → Keep context nearby: Note the fixed tasks and absence of a significance test.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Zero-based correct counts



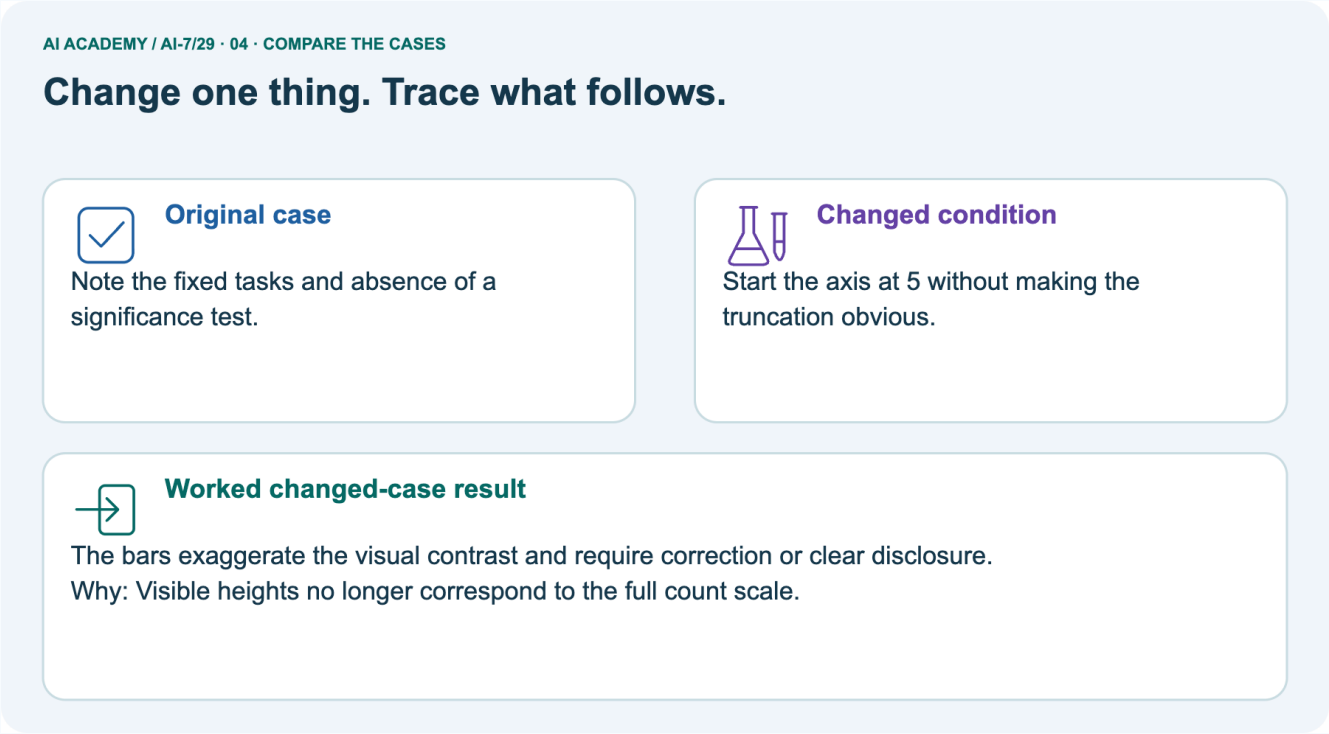
Follow the labelled picture with the matching explanation. Open the vector diagram

Zero-based correct counts. Units: correct cases out of 10. A: 6. B: 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

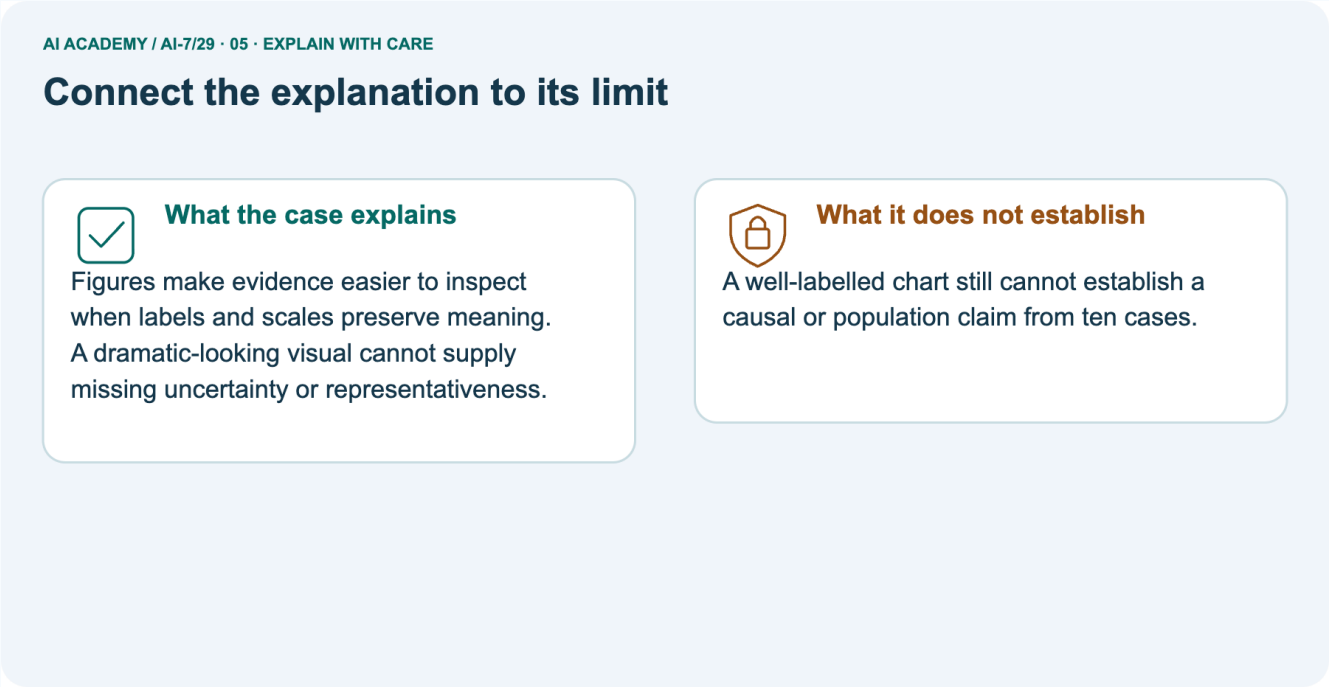
Original case: Note the fixed tasks and absence of a significance test. Changed condition: Start the axis at 5 without making the truncation obvious.

Expected result: The bars exaggerate the visual contrast and require correction or clear disclosure. Reason: Visible heights no longer correspond to the full count scale.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

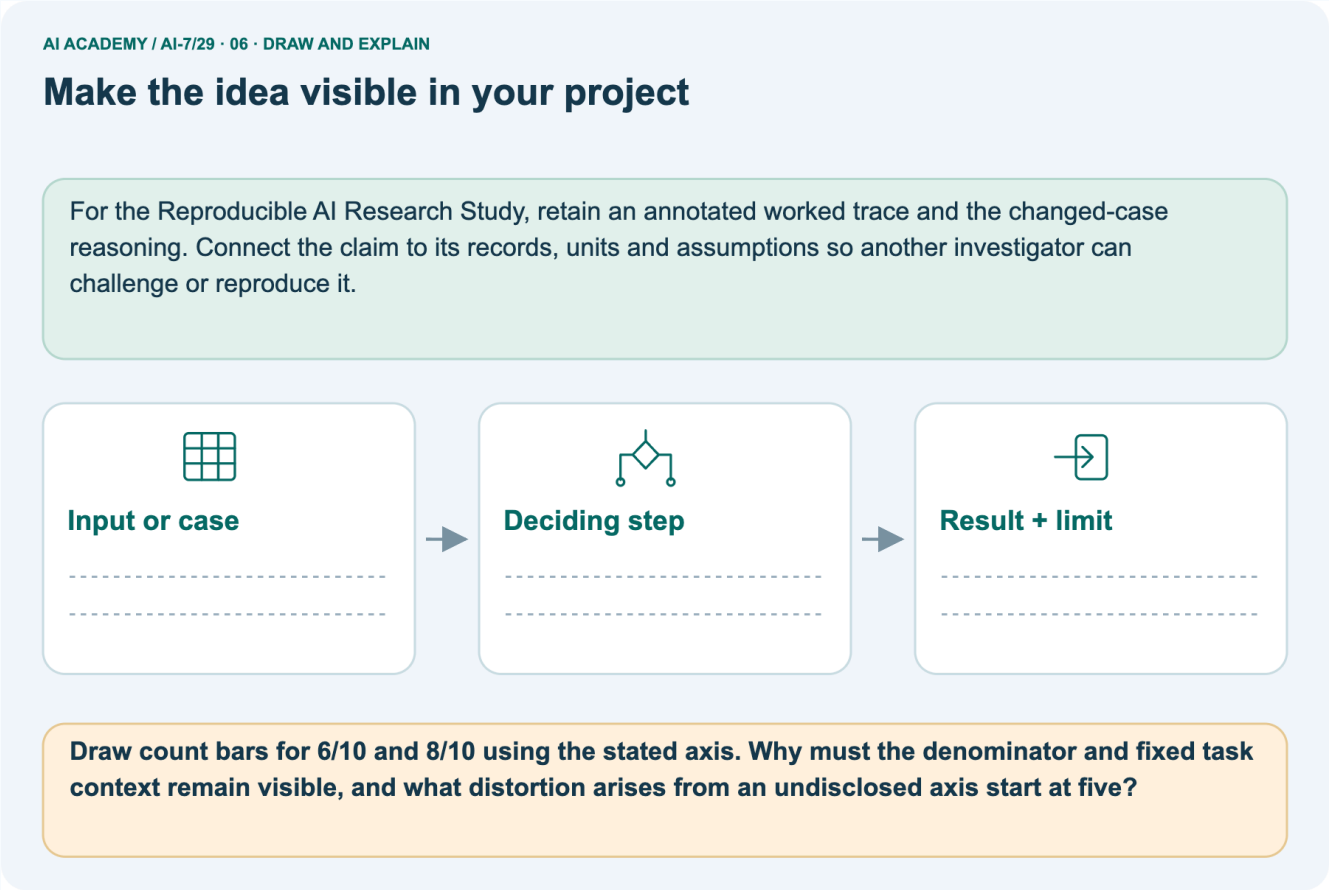
Figures make **evidence** easier to inspect when labels and scales preserve meaning. A dramatic-looking visual cannot supply missing uncertainty or representativeness.

Limit: A well-labelled chart still cannot establish a causal or population claim from ten cases.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Draw count bars for 6/10 and 8/10 using the stated axis.

Why must the denominator and fixed task context remain visible, and what distortion arises from an undisclosed axis start at five?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Figures that preserve evidence: the case

02. Follow the deciding steps

03. Zero-based correct counts

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.